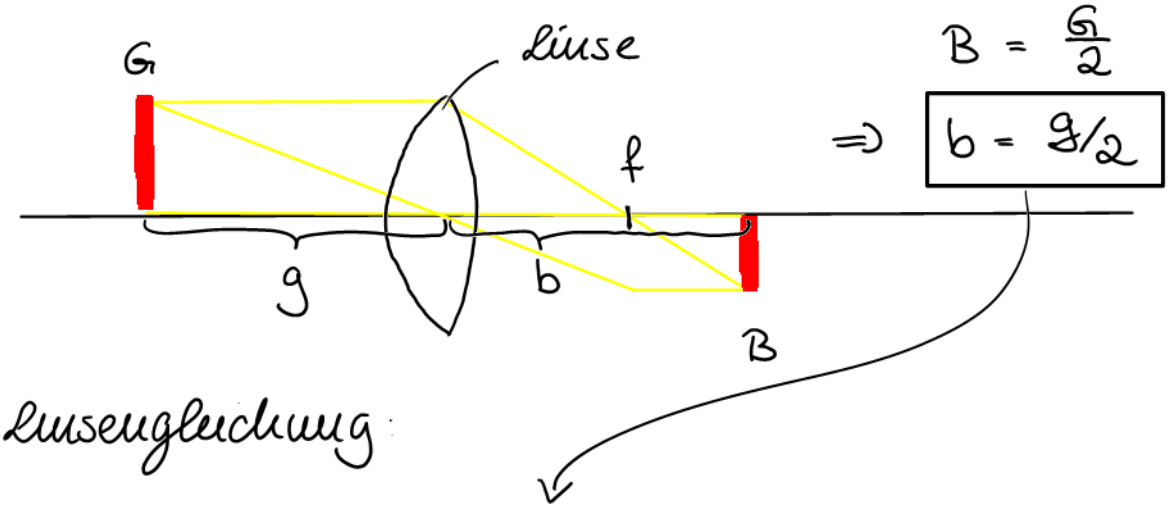
Musterlösung - Bild und Gegenstandsgröße



$B = \frac{G}{2}$

$\Rightarrow \boxed{b = g/2}$

Linsengleichung:

$$\frac{1}{b} + \frac{1}{g} = \frac{1}{f} \Rightarrow \frac{2}{g} + \frac{1}{g} = \frac{1}{f}$$

$$\Rightarrow \frac{3}{g} = \frac{1}{f} \Rightarrow \underline{\underline{f = g/3}}$$

Brennweite aus Linsenradien:

$$\frac{1}{f} = (n-1)\left(\frac{1}{r_1} + \frac{1}{r_2}\right)$$

$$\approx (1{,}5-1)\left(\frac{1}{30\,cm} + \frac{1}{50\,cm}\right) = 0{,}5 \cdot 5{,}\overline{3} \cdot 10^{-2}\,cm^{-1}$$

$$\underline{\underline{f = 37{,}5\,cm}}$$

$$\Leftrightarrow \quad g = 3f = \underline{\underline{112{,}5\,cm}}$$

$$b = g/2 = \underline{\underline{56{,}25\,cm}}$$